Potential Dividers

Exam code: 0625 & 0972

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Written by: Katie M
Reviewed by: Caroline Carroll
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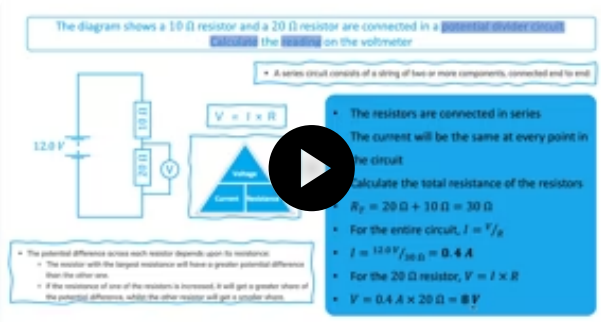

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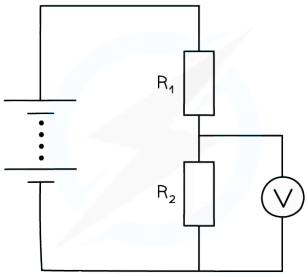
Variable potential dividers

Extended Tier Only

- When two resistors are connected in series, the potential difference across the power source is shared between them
- A potential divider is a circuit which **splits** potential difference from a power source, so only a **fraction** goes to a component (a voltmeter, in the diagram below)

Potential divider diagram



A potential divider splits the potential difference of a power source between two components

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

greater potential difference than the other one
    - If the **resistance** of one of the resistors is **increased**, it will get a **greater** share of the potential difference, whilst the **other** resistor will get a **smaller** share

- If one resistor is a **variable resistor**, the potential difference across the other resistor can be **altered**
    - This means the potential difference across any component **in parallel** with that resistor can also be altered

Resistors as Potential Dividers

Extended Tier Only

- When two resistors are connected in series, the power source's e.m.f. is split between the resistors
    - This potential difference splits in the **same ratio** as the resistance of the two resistors

- The ratio of potential differences across each resistor can be found using the following equation:

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

- Where:
    - R_1 is the resistance of resistor 1 in ohms, Ω
    - R_2 is the resistance of resistor 2 in ohms, Ω
    - V_1 is the potential difference across resistor 1 in volts, V
    - V_2 is the potential difference across resistor 2 in volts, V

- Recall that the e.m.f. of the power source will be equal to $V_1 + V_2$

Worked Example

The circuit is designed to light up a lamp when the input voltage exceeds a value.

When the lamp lights up, V_{out} is 5.3 V.

Calculate the e.m.f. of the power source required to illuminate the lamp.

Answer:

Step 1: List the known quantities

- Resistance of resistor 1, R_1 = 12 kΩ = 12 000 Ω
- Resistance of resistor 2, R_2 = 20 kΩ = 20 000 Ω
- Potential difference across resistor 2, $V_2 = V_{out}$ = 5.3 V

Step 2: Recall the equation for a potential divider

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

Step 3: Substitute the known quantities and determine the potential difference across resistor 1

$$\frac{12\,000}{20\,000} = \frac{V_1}{5.3}$$

$$V_1 = 3.18 \text{ V}$$

Step 4: Determine the e.m.f. of the power source, V_{in}

$$V_{in} = V_1 + V_2$$

$$V_{in} = 3.18 + 5.3 = 8.48 \text{ V}$$

- The e.m.f. of the power source when the lamp illuminates is 8.5 V to 2 significant figures

Examiner Tips and Tricks

When thinking about potential dividers, remember that the higher the resistance the more energy it will take to 'push the current through' and therefore the higher the potential difference.

This means that if a component (often shown as a voltmeter in questions) needs to be switched on by a change such as increased light or temperature, then the resistor it is in parallel with needs to become larger **compared to the other resistor.**

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